Year II — Algebra Problem Set 1-12

Based on lectures by Various Lecturers

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Chapter 1

Rings, Subrings and Fields

Question 4

Consider the set of polynomials $\mathbb{Q}[x]$ with its normal addition, but instead of the normal multiplication, define a new binary operation: For any $f, g \in \mathbb{Q}[x]$ let $f \circ g$ be the polynomial obtained by substituting $g(x)$ into $f(x)$, that is,

$$(f \circ g)(x) = f(g(x)).$$

To check that $f \circ g$ is indeed a polynomial, write $f(x) = a_0 + \cdots + a_n x^n$ and $g(x) = b_0 + \cdots + b_m x^m$. We can then expand $f(g(x))$ using the multinomial theorem (i.e., repeated use of the binomial theorem), to see that $f(g(x)) \in \mathbb{Q}[x]$. Does $\mathbb{Q}[x]$ equipped with the operations $+$ and $\cdot$ form a ring?

Proof. To form a ring we must first ensure that the $\mathbb{Q}[x]$ forms an Abelian Group with respect to the $+$ operation. We note that addition forms an Abelian group as it operates the same as real number addition. We now check the remaining three ring properties.

1.0.1 Identity

We wish to find $e(x)$ such that:

$$(f \circ e)(x) = (e \circ f)(x) = f(x)$$

Since $e(x) \in \mathbb{Q}[x]$ this must be a polynomial. Subbing any other polynomial into this polynomial must leave it unchanged and we can sub it into any polynomial to leave it unchanged. Our identity is therefore $e(x) = x$.

1.0.2 Associativity

Next we want to conclude:

$$f(x) \circ (g \circ h)(x) = (f \circ g)(x) \circ h(x)$$

Dealing with the LHS:

$$f(x) \circ (g \circ h)(x) = f(x) \circ g(h(x)) = f(g(h(x)))$$

and then the RHS

$$(f \circ g)(x) \circ h(x) = f(g(x)) \circ h(x) f(g(h(x)))$$

Therefore we have associativity.

1.0.3 Distributivity

Finally we want to check if:

$$f \circ (g + h) = f \circ g + f \circ g$$

However, starting with the LHS we have:

$$f \circ (g + h) = f((g + h)(x)) = f(g(x) + h(x))$$

And on the RHS:

$$f \circ g + f \circ h = f(g(x)) + f(h(x))$$

These are not directly equal in all cases. For instance, consider $f(x) = x^2, g(x) = x, h(x) = x$. On our LHS we produce:

$$f \circ (g + h) = 4x^2$$

but on our right we produce:

$$f \circ g + f \circ h = 2x^2$$

Therefore $(\mathbb{Q}[x], +, \circ)$ is not a ring.

□